Project 6

AI-Driven Intelligent Academic Assessment and Performance Analytics System

Agentic AI-Based Academic Evaluation & Learning Analytics Platform (SIGER)

Problem Statement

Design and implement an Agentic AI-based Intelligent Academic Assessment and Performance Analytics System that automates the evaluation of student submissions while providing comprehensive learning analytics for both students and educators. The system should support secure role-based access through dedicated student and faculty interfaces, enabling students to upload assignments, track submission status, view grades, and receive personalized feedback, while allowing faculty to evaluate submissions, monitor assessment completion, identify learning gaps, and analyze subject-wise student performance. The system should integrate automated assessment evaluation, Retrieval-Augmented Generation (RAG)-based feedback generation, LLM-based performance analysis, and real-time analytics to deliver explainable, data-driven insights that improve teaching effectiveness, student learning outcomes, and academic decision-making.

Objective 1 — Intelligent Automated Assessment and Personalized Feedback

- Develop an agent that automatically evaluates student assignments, projects, and assessments to determine topic-wise mastery and overall subject understanding.
- Integrate LLM-based evaluation, RAG-based knowledge retrieval, and automated feedback generation to provide personalized, explainable, and constructive academic feedback.
- Compute subject-wise comprehension scores, grading metrics, and concept mastery levels to support continuous student learning.

Objective 2 — Dual-Role Academic Analytics and Performance Monitoring

- Develop secure Student and Faculty portals with role-based access for managing submissions, assessments, grades, and performance reports.
- Enable faculty to monitor submission completion, overdue assessments, learning gaps, and aggregated class-level performance analytics through real-time dashboards.
- Provide students with submission tracking, grade visualization, personalized progress reports, and continuous performance monitoring, while ensuring data security, transparency, reliability, and explainability throughout the academic assessment process.

System Architecture — End to End

The diagram below shows the complete flow of the system, from the Student and Faculty portals down to the underlying data and AI model layer. The design keeps orchestration, support services, and storage clearly separated so each part stays simple and explainable.

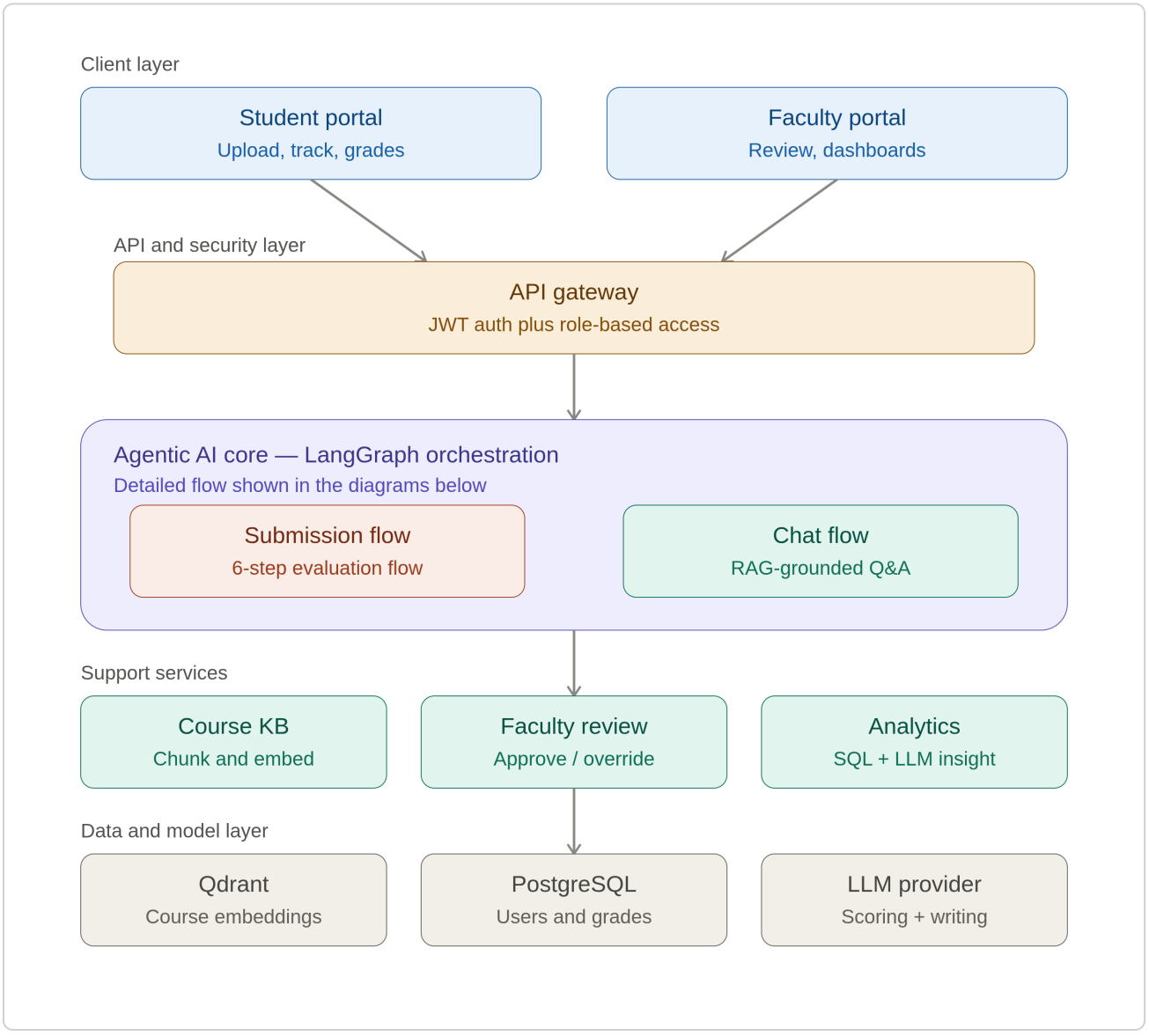


Figure 1: Layered end-to-end architecture — Client, API/Security, Agentic AI Core (LangGraph), Support Services, and Data/Model layers.

Submission Flow — Detailed View

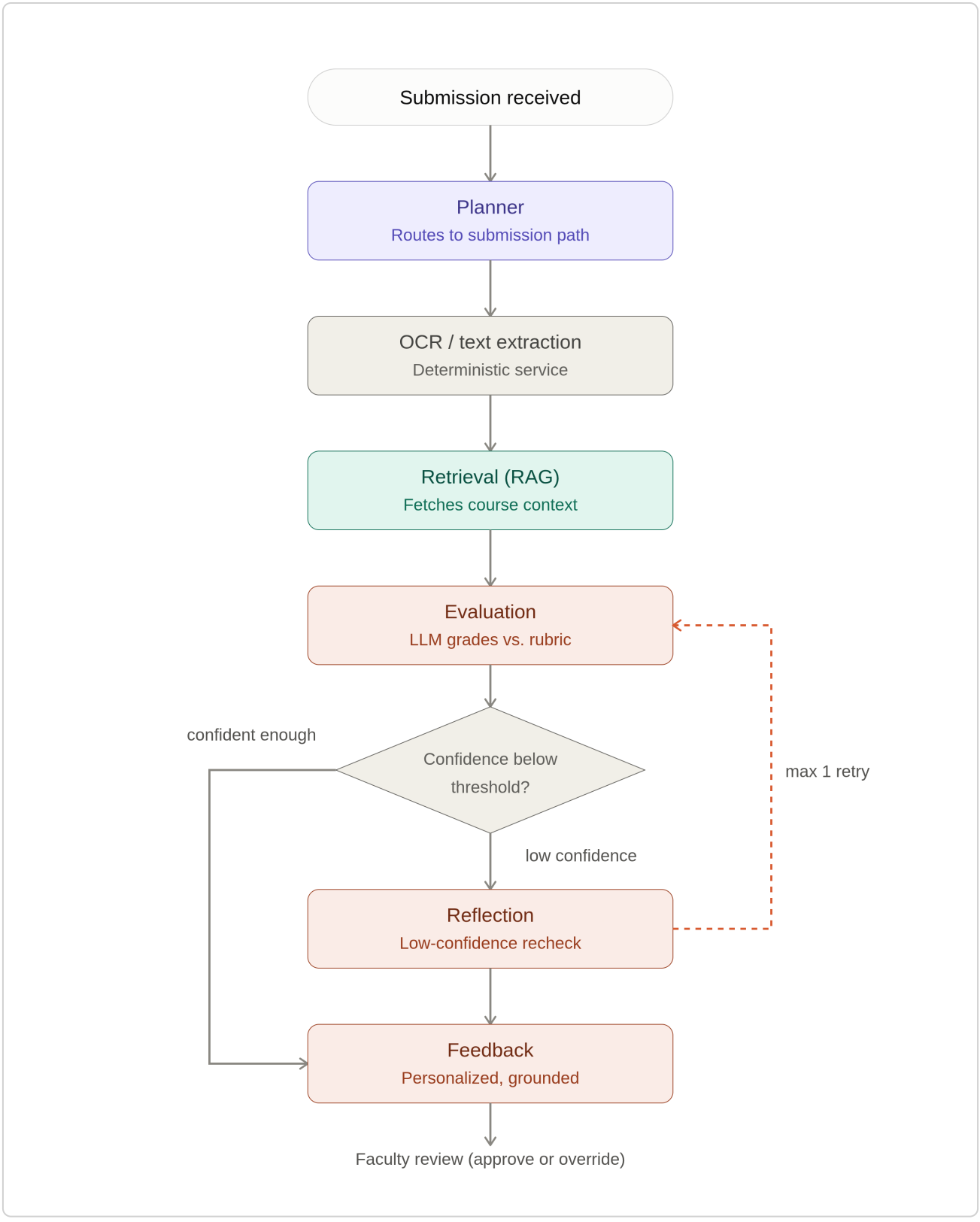


Figure 2: Step-by-step submission evaluation flow — Planner, OCR, Retrieval, Evaluation, a capped low-confidence Reflection retry, Feedback, and Faculty review.

Chat / Teaching Flow — Detailed View

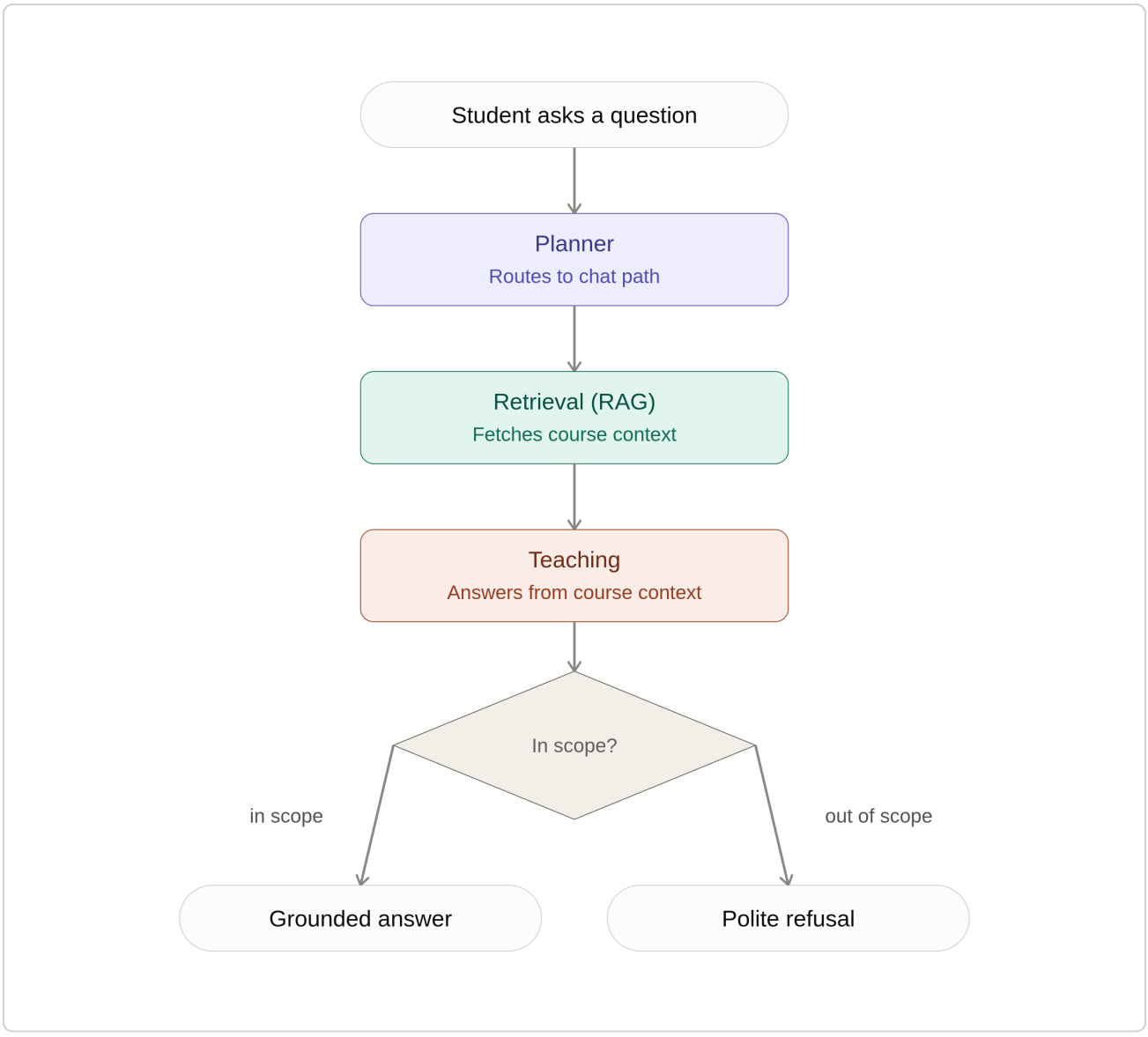


Figure 3: Student Q&A flow — Planner and Retrieval feed the Teaching node, which gives a grounded answer or a polite refusal for out-of-scope questions.

How the Flows Work

Submission Flow (Automated Assessment)

Student Portal → API Gateway (JWT/RBAC) → Planner → OCR/Text Extraction → Retrieval (RAG) → Evaluation (LLM grading) → Reflection (only if confidence is low, capped retries) → Feedback (personalized, grounded in course material) → Faculty Review (approve/override) → Final grade published back to the Student Portal.

Chat / Teaching Flow

Student Portal → API Gateway → Planner → Retrieval (RAG) → Teaching node answers academic questions using retrieved course context, with an out-of-scope refusal for unrelated queries.

Performance Analytics Flow

PostgreSQL data → deterministic SQL aggregation (scores, completion, topic performance, trends, rankings) → LLM Performance Analysis interprets the computed metrics into explainable insights and focus areas (the LLM never invents the raw numbers) → results surfaced on Faculty and Student dashboards.

Design Principles

- **Separation of concerns:** deterministic services (OCR, SQL analytics) are kept separate from LLM reasoning (evaluation, interpretation, feedback).
- **Human-in-the-loop:** faculty always reviews and can override an AI-generated grade before it is published.
- **Explainability:** every evaluation carries a score, rationale, topic, and confidence value; every analytics insight is grounded in SQL-computed metrics.
- **Security:** JWT authentication with strict student/faculty role-based access control across every endpoint.